

MURPHY CHAUDHARY

Electrical Engineer

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PROFESSIONAL SUMMARY

Electrical Engineering graduate with a strong technical foundation in power systems analysis, control engineering, and advanced simulation tools. Proven capability in integrating machine learning and software development with traditional electrical engineering concepts to design, model, and optimize complex systems. Adept at applying analytical problem-solving skills to electrical installation design, low-voltage distribution, and real-time computational modeling while adhering to industry standards.

SKILLS

- Core Engineering:** Power System Analysis & Design , Electrical Installation Design , High Voltage Engineering , Control Engineering
- Software & Programming:** MATLAB & Simulink (Advanced) , Python (Data Science & Machine Learning), C/C++, Office tools
- Standards & Tools:** NEC/IEC Standards , Load Schedules, Single Line Diagrams (SLD)

ACADEMIC PROJECTS

Spiking Neural Network (SNN) Controlled SAPF

December 2025 – May 2026

- Designed and simulated a Shunt Active Power Filter (SAPF) in MATLAB/Simulink to dynamically mitigate grid harmonics.
- Developed an event-driven Spiking Neural Network (SNN) controller in Python to replace traditional, slower numerical solvers.
- Achieved low latency and real-time calculation of compensating currents to drastically reduce Total Harmonic Distortion (THD).

Illumination & Low Voltage Distribution System Design

August 2025 – November 2025

- Designed a comprehensive interior/exterior lighting layout and low-voltage electrical distribution system matching NEC/IEC standards.
- Performed lux level calculations and lumen methods to optimize fixture placement and minimize energy consumption.
- Produced detailed electrical installation schematics, load schedules, and single-line diagrams (SLD).

Transmission Line Design App

August 2025 – November 2025

- Modeled electrical line parameters (R, L, C), transient phenomena (Ferranti Effect), and voltage regulation under heavy inductive grid loads.
- Developed and deployed a public web application using Python and Streamlit to automate complex geometric and electrical calculations.
- Integrated dynamic user interfaces allowing instant visualization of transmission efficiency and parameter variations.

EDUCATION

Bachelor in Electrical Engineering

2022-2026

Pulchowk Campus, IOE

REFERENCES

Asst. Prof. Anil Panjiyar
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